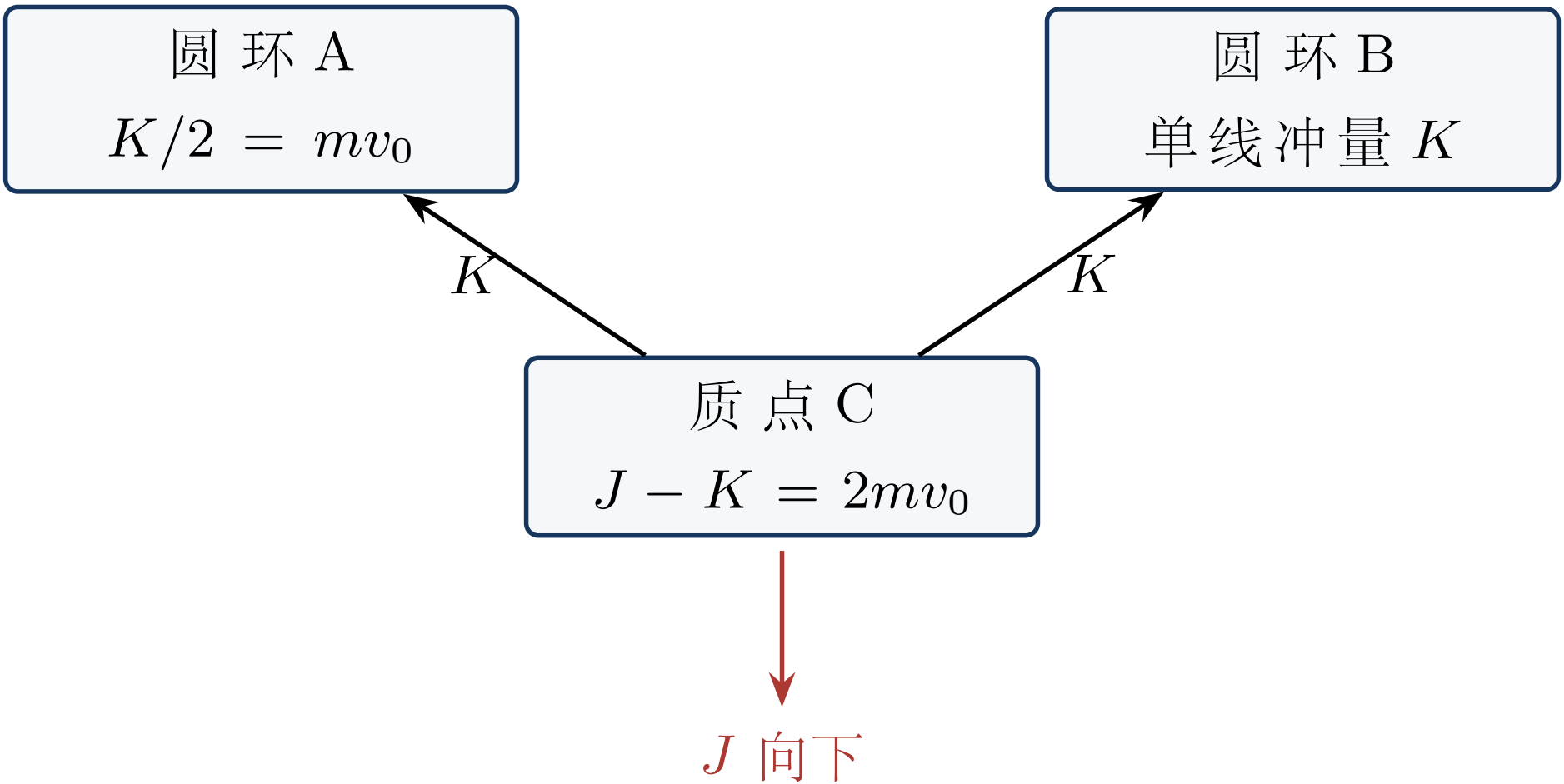
圆环沿杆速度 v_0；由 $K/2 = mv_0$

得 $K = 2mv_0$，外冲量 $J = 4mv_0$。